33. Scatter Plot with Empty Circles

Aim

To write a Python program to draw a scatter plot using random distributions with empty circles.

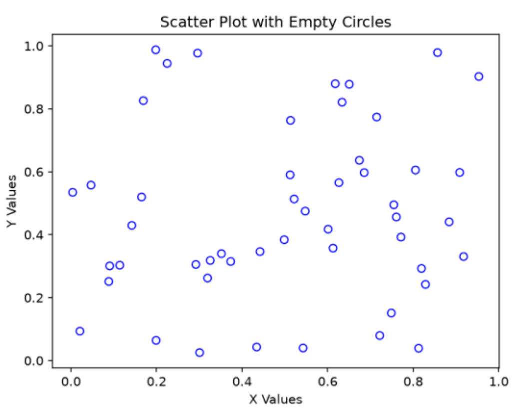
Dataset

Random X and Y values are generated using NumPy.

Python code:

```python
import matplotlib.pyplot as plt
import numpy as np
np.random.seed(10)
x = np.random.rand(50)
y = np.random.rand(50)
plt.scatter(x, y, facecolors="none", edgecolors="blue")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Scatter Plot with Empty Circles")
plt.show()
```

Output:



Result:

A scatter plot is displayed with randomly distributed **empty circle markers**.

34. Scatter Plot with Balls of Different Sizes

Aim

To write a Python program to create a scatter plot using random distributions and represent the points with different sizes.

Dataset

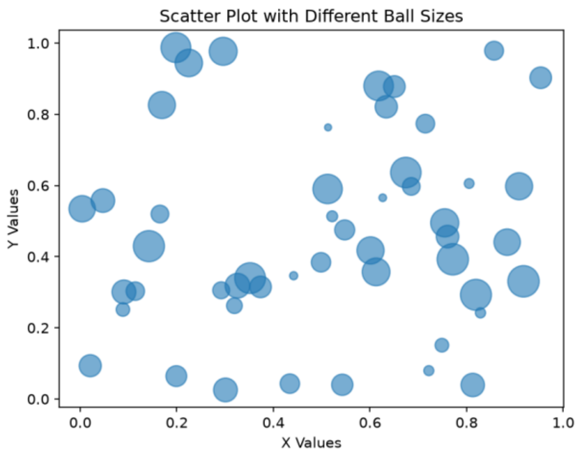
Random X, Y, and size values are generated.

X	Y	Size
Random	Random	Random
Random	Random	Random
Random	Random	Random
Random	Random	Random

Python

```
import matplotlib.pyplot as plt
import numpy as np
np.random.seed(10)
x = np.random.rand(50)
y = np.random.rand(50)
sizes = np.random.randint(20, 500, 50)
plt.scatter(x, y, s=sizes, alpha=0.6)
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Scatter Plot with Different Ball Sizes")
plt.show()
```

Output:



Result:

A scatter plot with randomly generated points of different sizes was successfully created.